Riley Beenders

ELECTROMECHANICAL & CONTROLS ENGINEER | INDUSTRIAL AUTOMATION | RELIABILITY

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PROFESSIONAL SUMMARY

Electromechanical engineer who has designed, wired, and troubleshot industrial machines and their controls in a manufacturing plant since 2020, and designs each revision to be more reliable and easier to service. Works independently across mechanical, electrical, and software disciplines and documents every system for the people who support it next.

CORE EXPERTISE

Controls & Automation: Modbus TCP data acquisition; multi-board CAN-bus control architectures; Siemens PLC support on production lines; servo, linear-motor, and braked NEMA 34 stepper motion; sensor and I/O integration; control-system performance monitoring and alarming; Allen-Bradley (Rockwell) PLCs

Electrical Design & Documentation: Electrical layout and schematic design to IEEE conventions; AutoCAD Electrical, KiCad, Altium, and Visio; wiring harnesses, interconnects, and cut-panel design; 120/240 VAC machine power; engineering documentation and BOM control; SolidWorks, Autodesk Inventor, Siemens NX, CATIA, Fusion 360

Software & Scripting: Python (asyncio, SQLite); LabVIEW; TypeScript and JavaScript (Electron, Node.js); PowerShell; MATLAB, Simulink, and Simscape; Git and GitHub; Linux (Debian) and Windows administration; VLAN segmentation of production networks

Safety, Reliability & Leadership: Risk and hazard analysis and mitigation; product safety and regulatory compliance; root-cause troubleshooting; vendor and fabrication coordination; project budgeting and status reporting; Lean Six Sigma; hiring and technical interviews; worldwide project collaboration

PROFESSIONAL EXPERIENCE

Proteor | Coplay, PA — January 2025 - Present

Lead, Research & Development Engineer

Proteor acquired Filament Innovations in January 2025, and the CTO role below carried forward as Lead R&D Engineer for the same product line, now sold as Proteor Print.

- Serve as the technical subject-matter expert across the plant, from industrial electronics and Siemens PLCs on the three filament extrusion lines to low-voltage sensors, and as the escalation contact for customer troubleshooting across mechanical faults, electrical faults, and user error.
- Built ExtrusionStudio, the plant's measurement and QC system, live on all three filament extrusion lines since October 2026 with no unplanned outage during production. A LabVIEW proof of concept established Modbus TCP communication with each line's tri-axial laser gauge; a Python rewrite then replaced three single-line executables with one service that polls every line concurrently at 20 Hz, so a powered-down line never stalls the others.
- Monitor every line against its tolerance band: an offset for the 1-2 minute lag between the laser and the spool ties each reading and out-of-spec alarm to the filament being wound, and a line that goes out of spec turns red on the floor projector with a timer showing how long it has been out.
- Gave every spool a serial, CSV record, QR label, and certificate, so the packing bench approves or rejects spools against gauge data instead of by eye; print-on-demand thermal labels reduced label waste by up to 30% per run.
- Resolved packing-bench panel lag and dropped connections at go-live by ruling out causes one at a time (frame rate, Raspberry Pi thermals, a fresh Pi and SD card, the network, display scaling): a hard-wired Ethernet run fixed the connection and the label-printer delay, and a display-scaling change removed the lag.
- Cut assembly time by 80% compared to previous designs on ICARUS-Lite, a compact industrial printer that automates prosthetic and orthotic production; led its design and development and owned the mechanical, electrical, and software design, including custom AC heaters and an IEEE-style electrical design adaptable to 120 or 240 VAC input.
- Completed product changes supporting conformity with the 2006/42/EC Machinery Directive and performed risk analysis and mitigations to EU and U.S. standards across ProteorPrint hardware and cloud products; ICARUS-Lite and ICARUS-Pro are now in CE testing.
- Re-engineered POSEIDON and ICARUS around one linear-motor motion platform with universal mounts, replacing servo-driven ballscrews to reduce product-to-product complexity and simplify service; validated 4-micron precision at 4 m/s on a 30 kg moving mass.
- Co-developed a custom hotend with an outside manufacturing partner, improving reliability and reducing cost for the company's materials.
- Contributed machine requirements and integration to ProteorPrintOS, built with external partner SimplyPrint, enabling remote operation and AI-assisted failure detection with live video and material, waste, and outcome metrics.
- Developing an offline version of the ICARUS-Lite and ICARUS-Pro control screen, together with over-the-air automated updates, so customers can run without a Wi-Fi or Ethernet connection.

- Ran the Coplay site's five-week Microsoft Business Central ERP implementation (2,000+ items, 150+ products) and the 2026 inventory count, tracing discrepancies to unverified production orders and helping move to a warehouse-focused workflow.
- Estimate and track the budget of every engineering project, build its BOMs, and maintain each product's unit cost; present status, constraints, and budgets at Proteor's monthly USA review, write incident reports for leadership, and track work on GitHub and Kanban boards in Microsoft Teams.
- Act as Hiring Manager for engineering roles, conducting technical interviews and owning final hiring decisions.

Filament Innovations | Whitehall, PA — May 2020 - January 2025

Also wrote customer proposals, scopes of work, and estimates.

Chief Technology Officer — January 2023 - January 2025

- Oversaw the product-development pipeline, internal IT infrastructure, and technology strategy while keeping hands-on ownership of mechanical, electrical, firmware, and software systems, and gave the CEO and the company regular updates as designs and software iterated.
- Led the design and launch of Gen3 POSEIDON, combining a 1 m^3 build volume, filament-and-pellet dual extrusion, industrial-servo XY motion, and a custom gantry.
- Designed and hosted a redundant Kubernetes cluster running a self-developed ERP/CRM for orders, inventory, tasks, projects, and customer interactions, with dynamic BOM generation linked to purchasing.
- Researched high-speed linear motors that moved a 30 kg payload at 3 m/s with 5-micron accuracy, and conducted early 6-axis robotic non-planar printing research.

Lead Electro-Mechanical Engineer — June 2022 - January 2023

- Simplified wiring and debugging on larger, modular machines by integrating a multi-board CAN-bus control architecture.
- Developed independently testable electronics enclosures that install in minutes, and coordinated custom harnesses, cut-panel designs, and hardware improvements with vendors.
- Improved thermal and acoustic performance through enclosure redesign and component placement, including custom tooling with local fabricators for large-radius aluminum-panel bends.

Industrial Engineer — April 2021 - June 2022

- Expanded motion safety on next-generation ICARUS and POSEIDON models with higher-performance components, NEMA 34 motors, planetary gearboxes, and power-failure brakes that prevent uncontrolled axis motion during power loss.
- Designed and released Kratos, a machine engineered to print high-strength above-knee and below-knee prosthetic sockets.
- Designed THE KRAKEN, a 1 x 2 x 1.5 m hybrid pellet-and-filament gantry machine for a client producing the world's largest 3D-printed statue, and managed CAD, BOM, sourcing, fabrication, assembly, firmware tuning, customer support, and operator training.

Engineering Technician — May 2020 - April 2021

- Improved early product reliability by diagnosing mechanical and electrical problems during assembly and operation.
- Tuned, calibrated, and optimized machine parameters while documenting design limitations and opportunities for product improvement.

ADDITIONAL RELEVANT EXPERIENCE

Blue Mountain Ski Resort - Ski Lift Operator (seasonal) | Palmerton, PA | November 2019 - January 2021 | Performed motor and control safety checks on passenger ski lifts and supported the maintenance team during lift breakdowns, while operating lifts and assisting thousands of guests.

Northampton Area School District - Computer / Network Technician | June 2016 - August 2019 | Built network infrastructure and deployed more than 750 wireless access points across six school buildings; accelerated deployment of more than 2,500 Chromebooks with an Arduino-based tool that automated setup-data entry, then coordinated final device distribution with school administrators.

RJB.Engineering - Owner & Engineering Consultant | 2015 - Present | Deliver project-based engineering support for automotive clients and operate a small online store for printer heat inserts and accessories; produced plastic face shields during the COVID-19 pandemic.

EDUCATION & CERTIFICATIONS

Penn State University | Bachelor of Science in Electromechanical Engineering — December 2024

MATLAB, Simulink, Simscape, and Machine Learning Onramps, MathWorks — 2026

Lean Six Sigma Yellow Belt, Educate 360 — 2025

Lean Six Sigma White Belt, Educate 360 — 2025

Certificate in Engineering Design and Tools, Penn State University — 2023

Mechanical Engineering Certificate, Lafayette College — 2019

SELECTED PROJECTS

BulkSSH | Windows desktop app, open source at github.com/RileyBeenders/BulkSSH October 2026 - Present

TypeScript, Electron, Node.js, SSH

Problem: Testing ExtrusionStudio's Pi kiosks meant repeating the same commands on every Pi, one SSH window at a time, after every change to the image.

- Opens every device in a grid of live terminals, mirrors keystrokes to every ticked device, and broadcasts a command with each device's output and exit code, so a change to the Pi image is tried on every Pi in one pass.
- Requires confirmation before risky commands such as reboot or rm -r reach several devices, never accepts host keys silently, and reconnects dropped connections by itself; released as an installer and a portable executable.
- Simulates SSH devices for development, so the app can be tried without hardware; saved passwords are optional and encrypted with the Windows account.

Production Network Segmentation | Ubiquiti UniFi, three independent sites March 2024 - Present

Problem: The plant's consumer-grade mesh network could not keep up as production grew: machines coming online together were sometimes assigned the same IP address, and the mesh did not detect or resolve the conflicts.

- Replaced it with UniFi cloud gateways, using VLANs to separate production equipment, work devices, and IoT devices, with central control of DHCP, DNS, port forwarding, and firewall routing.
- Reduced IP conflicts and network-related downtime, and uses monitoring tools to find and resolve connectivity problems faster.

RB's Auto App Updater | Windows utility, personal project September 2026

PowerShell, WinGet, Windows Task Scheduler, GitHub

- Updates every WinGet-managed application from one desktop shortcut, upgrading each package on its own so a single failure never stops the rest, and translates exit codes into plain-language statuses such as restart needed or application in use.
- Runs upgrades through an elevated scheduled task with no UAC prompt after setup, and updates itself from a GitHub version manifest only after the new script's SHA-256 hash matches; detailed logs make remote troubleshooting possible.